

# Disaster Nephrology

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## CHAPTER OUTLINE

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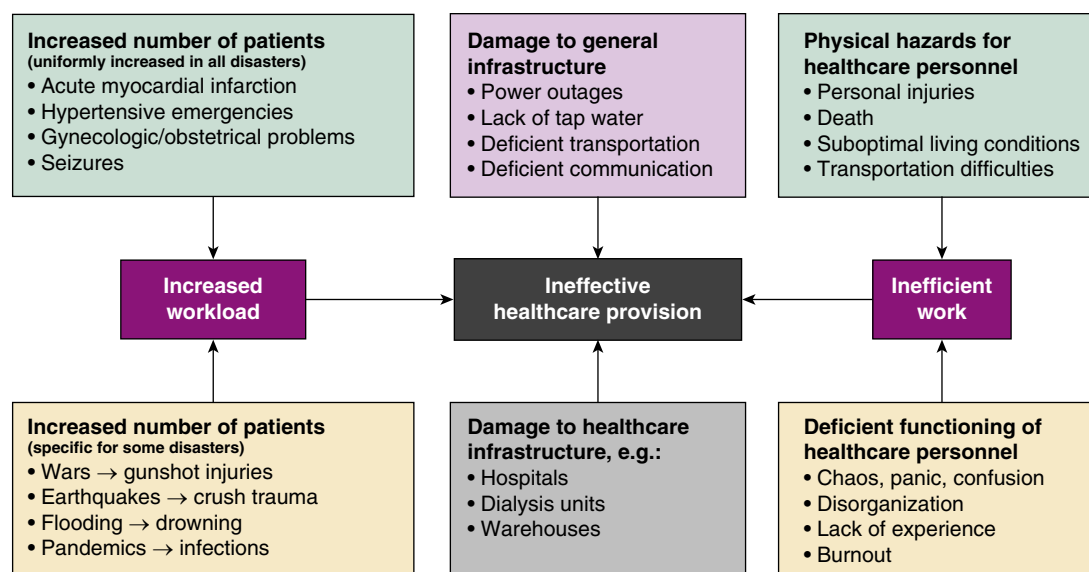
## KEY POINTS

- During massive disasters, the entire spectrum of patients with kidney diseases (i.e., those with acute kidney injury, chronic kidney disease not on kidney replacement therapy, and kidney failure on dialysis or with a functioning transplant) are at higher risk compared with the healthy population.
- Pediatric patients with kidney diseases, especially those with kidney failure, constitute an even higher risk group, especially if they are unaccompanied by caregivers.
- The only way to improve outcomes is to prepare for an effective response in the predisaster period at the regional, national, and international levels, which can minimize the extent of medical problems, as well as postdisaster chaos, confusion, and disorganization.
- All stakeholders (rescue team members, health care providers, authorities, patients, and their caregivers) should be trained on how to cope with disaster-related medical and logistic problems.
- Major steps in disaster response are implementation of predefined action plans, handling of immediate logistic problems, and management of patients with various kidney diseases.
- If the needs of patients with kidney diseases cannot be coped with locally, request for personnel and material support from outside becomes mandatory.
- When structural and infrastructural damage is extensive and provision of local health care becomes impossible, the only solution will be evacuation, which adds further risks for patients with kidney diseases.
- Professional societies should consider preparing guidelines for management of patients with kidney diseases in disaster conditions and providing education on disaster medicine, which differs considerably from routine medical practice.

## DISASTERS

Disasters, either natural or man-made, result in “serious disruptions of the functioning of a society associated with widespread individual, material, economic or environmental losses.”<sup>1</sup> Natural disasters (earthquakes, hurricanes, pandemics) arise from a sudden major

upheaval of nature, whereas man-made disasters (wars, terrorist attacks, nuclear disasters) develop as a consequence of damaging actions of humans. Disasters may be “destructive” (characterized by total or partial destruction of physical assets (e.g., in case of wars, earthquakes, and hurricanes) or “nondestructive” (which may result in extensive harm, however, with no structural



**Fig. 82.1 Reasons for ineffective health care provision during massive disasters.** When damage to general and health care infrastructure combines with shortage of skilled health care workforce, deficient health care is inevitable. An increased number of patients further increases these drawbacks (list of included factors in each box is not exhaustive). (From Sever MS, Pawlowicz-Szlarska E, Tuglular ZS. Nephrology in disasters, epidemics and wartime. In: Ronco C, Bellomo R, House A, et al, eds. *ERA Neph-Manual*. European Renal Association; 2024. Accessed 3 March 2025. <https://e-learning.era-online.org/ebook/chapter-22-nephrology-in-disasters-epidemics-and-wartime>.)

damage; e.g., during pandemics or droughts). Although a significant year-by-year variation exists, overall, there is a trend for an increase in the number of both natural<sup>2</sup> and man-made disasters over time,<sup>3,4</sup> at least in part related to political, cultural, and religious conflicts; growing world population; and an increase in mobility and environmental changes (e.g., global warming).

## MEDICAL PROBLEMS AFTER DISASTERS

Medical problems caused by disasters may be “disaster specific” (e.g., increased risk of crush syndrome after destructive events,<sup>5,6</sup> drowning after floods,<sup>7</sup> gunshot wounds during wars).<sup>8-10</sup> However, other medical threats increase uniformly with all types of disasters, such as the risk of developing acute myocardial infarction, stroke, seizures, or acute exacerbations of preexisting chronic diseases (e.g., hyperglycemia or ketoacidosis in diabetes,<sup>11</sup> uncontrolled hypertension,<sup>11,12</sup> and exacerbations in other chronic diseases)<sup>13,14</sup> due to disaster-related logistic drawbacks. Logistic difficulties contribute to the dimension of medical problems. When increased numbers of patients due to disaster-related or disaster-unrelated reasons are superimposed on damaged general and health care infrastructure, health care provision becomes ineffective. Substandard work of the health personnel will exaggerate these drawbacks (Fig. 82.1).<sup>15-17</sup>

## CHALLENGES IN PROVIDING KIDNEY CARE DURING DISASTERS

Overall, medical concerns are more prominent in vulnerable patients (i.e., elderly, frail, women, children, and all people with noncommunicable diseases). This is also the case for individuals with acute or chronic kidney disease due to many factors, including<sup>8,14,18-29</sup>:

- *Increased number of acute and chronic patients* per unit is a frequent epiphenomenon of destructive and nondestructive disasters.<sup>5,30,31</sup> For example, in the COVID-19 pandemic, acute kidney injury (AKI) incidence was 46% in all hospitalized victims; this figure increased to 76% in patients admitted to intensive care units (ICUs).<sup>32</sup> After the Marmara earthquake in 1999, 639 patients with AKI due to crush injuries were registered and 5137 sessions of hemodialysis were required in these patients.<sup>33</sup> Furthermore, due to movement of chronic patients from nonfunctioning dialysis units, there was also a relative increase in the workload in functional facilities.<sup>34,35</sup>
- *Damage to infrastructure* is a major problem for patients with kidney diseases. Those who need dialysis are exposed to the highest risk, arising from disruptions in water and electricity, difficulties with communication and transportation, and damage to hospital buildings and/or dialysis units/machines/materials, which all interfere with dialysis services.<sup>31,35-45</sup> Also, extensive chaos, panic, and disorganization, even in high-income and disaster-prone countries, may result in adverse outcomes.<sup>17,26,41,46</sup>
- *Shortage of specific therapeutics* is a major obstacle for patients with kidney diseases, who need many medications and supplies for treatment and survival. These items include but are not limited to peritoneal dialysis (PD) solutions, heparin, immunosuppressants, erythropoiesis-stimulating agents, antihypertensives, phosphate binders, and vitamin D and iron preparations.<sup>39,47</sup>
- *Shortage of well-trained personnel* (e.g., hemodialysis technicians, kidney nurses, nephrologists, and medical professionals involved with other specialties) is frequent during disasters due to personal injuries, transportation problems, concerns related with family members or personal property, and everyday family responsibilities.<sup>48,49</sup>

Concerningly, despite extensive literature,<sup>31</sup> awareness and knowledge about kidney problems during disasters remain disappointing among authorities, policymakers, rescuers,

the public, and most medical professionals. Therefore major shortcomings persist in health care provision because of not only disparity between health care demand and supply but also disaster-related specific problems in patients with various kidney diseases. Risk reduction and appropriate responses are extremely complex under these circumstances (Fig. 82.2).

### CHANGING CONCEPTS: FROM “RENAL DISASTER” TO “DISASTER NEPHROLOGY”

The link between massive disasters and a high number of patients with kidney disease was first recognized in the aftermath of the Spitak-Armenia earthquake in 1988, when 600 victims suffered from crush-related AKI.<sup>30</sup> Almost all of these victims died due to lack of dialysis support; subsequently, the term “renal disaster” was introduced and the Renal Disaster Relief Task Force of the International Society of Nephrology was created.<sup>50</sup> After the Marmara earthquake in Turkey in 1999, the concept of “seismonephrology” or “earthquake-nephrology” was developed, highlighting the frequent occurrence of AKI in the victims rescued from collapsed buildings.<sup>51</sup> Afterwards, however, considering the effects of Hurricane Katrina on patients with kidney diseases<sup>41</sup> and potential threats by other disasters (e.g., wars, tsunamis, pandemics) that may disrupt care of maintenance dialysis patients and individuals living with a functioning kidney transplant, the more encompassing term “disaster nephrology” was suggested.<sup>52,53</sup>

### SPECIFIC PROBLEMS IN PATIENTS WITH KIDNEY DISEASES

#### PATIENTS WITH AKI

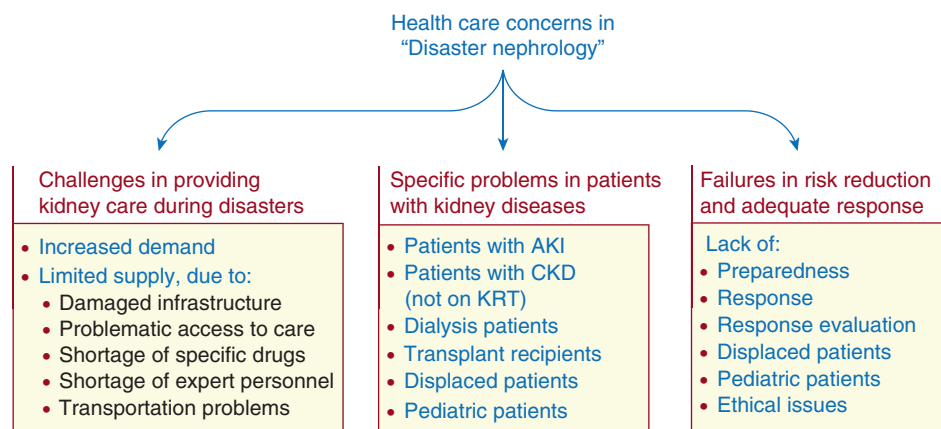
An increase in the number of patients with AKI due to different pathogenetic mechanisms has been observed in various disasters.<sup>54</sup>

#### EARTHQUAKES

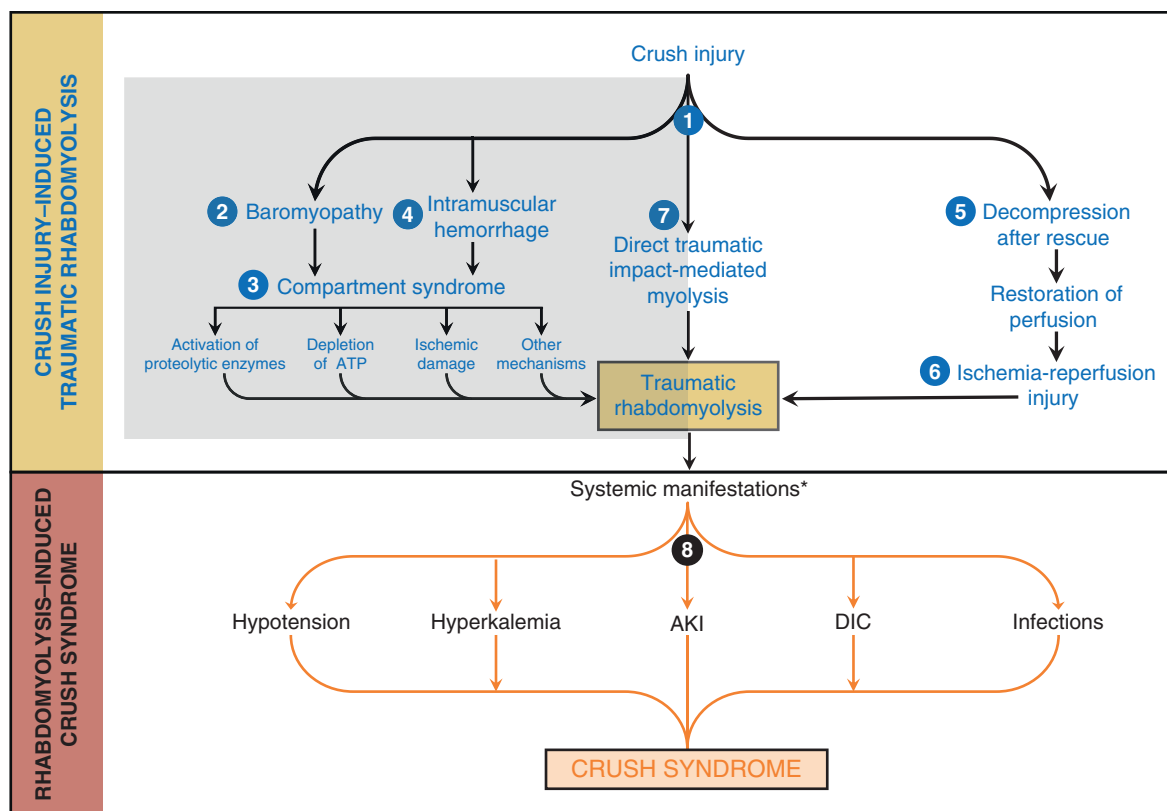
Several etiologies (e.g., bleeding-related hypoperfusion of the kidneys, transfusion reactions, dehydration after having been trapped in confined spaces, high incidence of infections and sepsis, exposure to nephrotoxic medications during treatment of complications) may play a role in the pathogenesis of earthquake-related AKI.<sup>31,54,55</sup> However, crush syndrome is the most frequent cause<sup>30,31</sup> and has been reported in up to 5% of all earthquake victims.<sup>5,56,57</sup> The first step in the pathogenesis of crush-related AKI is development of traumatic rhabdomyolysis in patients trapped under the rubble (Fig. 82.3).<sup>58-62</sup> This is followed by development of systemic responses forming the basis of the crush syndrome. For example, hypotension may develop due to hypovolemia secondary to fluid third-spacing in the muscle compartments; hyperkalemia may develop due to traumatic rhabdomyolysis and acidosis-induced efflux of potassium from the muscle cells; disseminated intravascular coagulation may be triggered by increased serum levels of thromboplastin released from injured muscles; and infections may result from contamination in traumatic/surgical wounds and unhygienic circumstances.<sup>5,56,57,63-65</sup> Many mechanisms may play a role in the development of crush-related AKI<sup>54,59-61</sup>: 1. intravascular volume depletion may result in renal hypoperfusion and ischemia; 2. muscular trauma increases serum myoglobin levels, which leads to myoglobinuria with subsequent intratubular cast formation and direct tubular toxicity of myoglobin; myoglobin scavenges nitric oxide and activates inflammatory pathways, all of which may contribute to kidney hypoperfusion and tissue injury; 3. high levels of serum uric acid originating from disintegrating cell nuclei may further contribute to cast formation and tubular obstruction; and 4. hyperkalemia and hypocalcemia-induced cardiac output depression may potentiate renal hypoperfusion.

#### WARS

The true incidence of AKI during wars is uncertain due to challenges with detection, diagnosis, and registration,



**Fig. 82.2 Spectrum of health care concerns in “disaster nephrology.”** During massive disasters, extra health care problems are added to routine demand, whereas health care provision is limited due to logistic reasons. Specific disaster-related problems in patients with kidney diseases exaggerate the dimensions of the crisis. Risk reduction by means of predisaster preparedness and response during the active phase of the disaster may only partially solve the problems. Retrospective evaluation of the response is mandatory to avoid similar mistakes in future disasters. Overall, drawbacks may be even more important in displaced patients and children. Of note, many ethical controversies exist in fair distribution of limited medical supplies and manpower. AKI, Acute kidney injury; CKD, chronic kidney disease; KRT, kidney replacement therapy.



**Fig. 82.3 Pathophysiology of traumatic rhabdomyolysis-induced crush syndrome.** Overall, pathogenesis may be divided into two major phases: crush injury–induced traumatic rhabdomyolysis and rhabdomyolysis–induced crush syndrome. Mechanisms leading to development of rhabdomyolysis take place when the victims are still under the rubble (gray shaded) or after being extricated, but these phases may overlap as well. 1. The initial event is crush injury, which may trigger pathogenesis when the victim is still under the rubble and continues after the extrication. 2. The major pathogenetic mechanism under the rubble is baromyopathy (or pressure-stretch myopathy), which increases sarcolemmal permeability. 3. Shift of plasma water into the muscle tissue triggers development of the compartment syndrome, whereas influx of calcium into cytosol activates proteolytic enzymes causing further myocyte and microvascular damage. Depletion of adenosine triphosphate (ATP) stores, ischemic damage, and many other mechanisms may also contribute to the pathogenesis of traumatic rhabdomyolysis. 4. Intramuscular bleeding may further increase intracompartmental pressure. 5. However, most of the damage occurs after the victim has been rescued. 6. Flow into the damaged tissue results in *ischemia-reperfusion injury*, which is one of the most important pathogenetic mechanisms. 7. Disruption of myocytes by direct effects of trauma plays a minor role. 8. Traumatic rhabdomyolysis triggers several systemic manifestations, such as hypotension, hyperkalemia, acute kidney injury (AKI), disseminated intravascular coagulation (DIC), infections, and many others resulting in crush syndrome. (Please see the text for details of rhabdomyolysis-induced crush syndrome development).<sup>58–62</sup> \*Systemic manifestations are not exhaustive.

although many factors may result in prerenal, intrinsic renal, and postrenal AKI. That is, prerenal AKI may develop due to gunshot injuries or penetrating trauma-induced bleeding and subsequent hypovolemia, and intrinsic AKI may occur on the basis of ischemic acute tubular necrosis due to prolonged prerenal states or myoglobinuric and hemoglobinuric acute tubular necrosis due to crush injuries and transfusion reactions, respectively.<sup>6,10,34,55,66</sup> Exposure to chemicals and airborne toxins leading to nephrotoxic acute tubular necrosis, sepsis-related AKI, and gunshot-related direct injury to the kidneys are other important causes of intrinsic AKI during conflicts. Postrenal AKI may result from urinary tract obstruction due to pelvic trauma and foreign bodies within the urinary tract.<sup>10,34</sup> These mechanisms apply similarly to terrorist attacks, which may involve gunshot injuries, explosions, and building collapses.<sup>67</sup>

#### OTHER DISASTERS

*Pandemics* drew the attention of the medical community during and after the COVID-19 disaster, when several reports defined a high incidence of AKI.<sup>32,38,68–73</sup> A detailed discussion of COVID-19–related AKI is provided in [Chapter 59](#).

AKI may also follow other man-made disasters such as traffic accidents and torture,<sup>74,75</sup> meteorologic disasters,<sup>7</sup> or malaria<sup>76</sup> and leptospirosis<sup>77</sup> during floods. However, data on these events are scarce.

Historically, outcomes of AKI after disasters remained dismal for many years, with mortality rates up to 41% in earthquake victims who needed dialysis support<sup>78</sup> and 53% in victims of wars.<sup>79</sup> Recently, these figures improved to nearly 20% among the recent earthquake-crush<sup>80–82</sup> and war-AKI victims.<sup>83</sup> Over the long term, AKI may result in chronic kidney disease (CKD). This adverse outcome has been reported in both war victims<sup>84</sup> and patients who had suffered from COVID-19,<sup>85</sup> although this has not received enough attention up to now.

#### PATIENTS WITH CHRONIC KIDNEY DISEASE NOT ON KIDNEY REPLACEMENT THERAPY

If patients can survive the acute phase, and if preparedness is adequate, CKD may remain stable during short-lived disasters. However, exceptions do exist (e.g., increased risk of rapid CKD progression and a significant increase in



dialysis initiation, especially in patients with hypertensive kidney disease, as experienced after the 2011 Great East Japan Earthquake and tsunami).<sup>86</sup> Standardized mortality rates for patients with CKD also increased in coastal areas after this earthquake.<sup>87</sup> Many factors (e.g., extreme stress, inability to obtain specific diet and medications, and shortage of experienced personnel) contribute to suboptimal treatment of underlying diseases<sup>8,16,27,31,41,47,66</sup> and consequent accelerated decline of kidney function. In protracted disasters, these risks are likely higher, although this has not been reported so far.

The management of people with CKD receiving immunosuppressants may be affected in two different ways. On the one hand, the immunosuppressive therapy may increase the risk of infection if they are subjected to unhygienic conditions, on the other hand not being able to continue their treatment may increase the risk for relapse of worsening of their primary disease.<sup>31</sup>

## HEMODIALYSIS PATIENTS

Across the whole spectrum of CKD, hemodialysis patients constitute the group at greatest risk during disasters, due to several reasons: 1. Shelling of dialysis centers, power outages, extreme weather, and infrastructural damage may cause disruptions of dialysis services.<sup>39,40,88,89</sup> 2. Dialysis needs may increase in maintenance patients because lack of appropriate diet, shortage of medications, and/or impossibility to obtain appropriate treatment, all of which may trigger complications.<sup>31,90,91</sup> 3. Increased incidence of AKI further amplifies dialysis shortage. 4. Patients may experience delays in arteriovenous fistula surgery due to increased need for other surgery or other reasons for increased pressure on the health care system.<sup>16,92</sup> 5. Work overload, burnout, and panic may result in suboptimal health care provision or even malpractice by the dialysis personnel.<sup>16,41,48</sup>

If an acute disaster (e.g., bombardment during wars or incident earthquakes) occurs when patients are connected to hemodialysis machines, patients themselves may attempt panicky disconnection, which increases the risks of extensive bleeding, subsequent hypotension, shock, and fistula problems.<sup>31,66</sup>

All these factors may result in missed dialysis sessions,<sup>35,38</sup> underdialysis, and, consequently, increased frequency of interdialytic and intradialytic complications,<sup>66,93,94</sup> which are all associated with higher odds of hospitalization<sup>95</sup> and mortality.<sup>16,89,96-98</sup> During disasters, high mortality rates have been reported among hemodialysis staff as well.<sup>99</sup>

## PERITONEAL DIALYSIS PATIENTS

The main challenge is obtaining dialysis solutions and other supplies because of damage to warehouses and pharmacies, as well as problems with delivery.<sup>100-104</sup> This drawback may contribute to inadequate dialysis and subsequent hypervolemia, electrolyte disturbances, heart failure, and hypertension.<sup>103,105</sup> Following destructive disasters, living in crowded shelters or camps<sup>26</sup> may increase the risks for peritonitis and exit-site and tunnel infections.<sup>43</sup> In patients on automated peritoneal dialysis (APD), two problems deserve special mention: first, lack of electricity

may interfere with its application;<sup>66</sup> second, panicky disconnection from APD machines at the moment of a disaster may lead to catheter damage, catheter dislocation, or peritoneal contamination.<sup>103</sup>

Despite all these drawbacks, outcomes during disasters among patients on PD are more favorable than for hemodialysis patients,<sup>106,107</sup> although, at least in the case of the COVID-19 pandemic, in some regions around the globe, high mortality rates were reported among patients on PD.<sup>104</sup>

## TRANSPLANT RECIPIENTS

Logistic challenges, inadequate nutrition, dehydration, shortage of immunosuppressive medications, failure to perform optimal follow-up, and unavailability of experienced medical personnel may all adversely affect outcomes of patients living with a functioning transplant.<sup>31,41,91,108-110</sup>

*Incident transplantation activity* is usually postponed during short-term disasters.<sup>41</sup> Declines in new transplantation numbers are more dramatic in protracted disasters, as is the case of long-lasting wars or pandemics.<sup>109,111,112</sup>

Most of the experience on the outcome of *previously transplanted patients* during disasters was reported after the COVID-19 disaster, during which people living with a functioning kidney transplant more frequently needed admission in the ICU<sup>113</sup> and had a higher mortality<sup>114</sup> than nontransplanted patients. The main reasons for these complicated courses and unfavorable outcomes were related to their immunosuppressed condition.<sup>115</sup> During a pandemic, minimizing immunosuppressive medications may decrease the risk of contracting the infection and its rapid progression, although this strategy may increase the possibility of rejection.<sup>116,117</sup>

## DISPLACED PEOPLE WITH KIDNEY DISEASES

Following a disaster, people may flee to another region in the same country (internally displaced people) or to another country (refugees). Risks faced by displaced patients with kidney diseases, especially by those with kidney failure, are more prominent than for the displaced healthy population both when traveling and in their new environment, because of disease-related and logistic drawbacks (i.e., more unhygienic and insecure conditions, need for dialysis, immunosuppressive state, and associated comorbidities).<sup>66,118-122</sup> Especially in the case of rushed displacement, life-threatening complications occur because of challenges in receiving appropriate medical care, reaching a dialysis unit, and inadequate communication of medical information.<sup>66</sup>

## PEDIATRIC PATIENTS

In natural disasters, up to 43% of acutely injured victims are children.<sup>123</sup> Every year, 175 million children around the world are estimated to be affected by natural disasters, and globally 449 million children—1 child out of 6—live in a conflict zone.<sup>124</sup> During disasters, the same general concerns observed in adult patients with kidney diseases also apply to pediatric kidney patients; however, differences may be noted with regard to a number of specific features (Table 82.1).<sup>5,16,125-131</sup> Children are at high risk of being

**Table 82.1** Differences in Various Features of Pediatric versus Adult Patients with Kidney Diseases During Disasters<sup>5,16,125-131</sup>

| Feature                                   | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overall health concerns                   | <ul style="list-style-type: none"><li>Malnutrition is more prominent in children</li><li>Children are more prone to communicable diseases in unhygienic environments</li><li>Breakdowns in vaccination programs are more frequent in children</li></ul>                                                                                                                                                                                                                          |
| Medical concerns related with kidney care | <ul style="list-style-type: none"><li>Prerenal AKI may be more frequent in children</li><li>Incidence of crush syndrome-related AKI is lower in the first decade of life</li><li>Prevalence of pediatric patients on KRT is lower than adult patients</li><li>There is less skilled pediatric health care personnel</li><li>Children are less compliant to dietary restrictions</li><li>Children are more susceptible to complications due to missed dialysis sessions</li></ul> |
| Logistic problems related to KRT patients | <ul style="list-style-type: none"><li>Obtaining pediatric-sized HD and PD equipment is more problematic</li><li>Unavailability of pediatric drug formulations is more frequent</li><li>Shortage of specialized pediatric dialysis personnel is more prominent</li></ul>                                                                                                                                                                                                          |
| General organizational problems           | <ul style="list-style-type: none"><li>Pediatric kidney patients are usually not included in preparedness plans</li><li>Many children are separated from their caregivers and are unaccompanied</li></ul>                                                                                                                                                                                                                                                                         |

AKI, Acute kidney injury; HD, hemodialysis; KRT, kidney replacement therapy; PD, peritoneal dialysis.

affected both physically and emotionally, because of limited self-preservation skills and dependence on their parents or caregivers.<sup>19</sup> Additionally, as compared with adults, children are more at risk of polytrauma and skull injuries, malnutrition, and dehydration because of their anatomic and physiologic characteristics.<sup>123,128,132-135</sup>

Harm may occur even in the prenatal period. Major concerns in that phase are intrauterine growth retardation, prematurity, and maternal mental disorders, which may cause long-lasting health problems including kidney damage.<sup>136,137</sup> However, despite all these extra risks, special needs of pediatric patients are often neglected in general disaster preparedness scenarios.<sup>138</sup>

**RISK REDUCTION AND RESPONSE**

Preparing communities and individuals appropriately for disasters in the *predisaster period* is vital to minimize postdisaster disorganization, panic, and chaos. This is extremely useful to prevent inappropriate reflexes and interventions during the *intradisaster period* and optimize

outcomes of the victims. In the *postdisaster period*, evaluation of the adequacy of preparations and the response is important to avoid repeating mistakes in future calamities (Fig. 82.4).<sup>31,34,139-141</sup>

**PREDISASTER PERIOD (DISASTER PREPAREDNESS)**

**ALL-HAZARDS APPROACH**

Different types of disasters (e.g., earthquakes, wars, hurricanes, pandemics, flooding) may cause similar practical problems (e.g., disorganization, panic, chaos, disruption of essential infrastructural services, shortages of housing and nutrition, interruptions in communication and transportation, disparity between health care demand and supply, unsecure environments).<sup>139,142-144</sup> However, demography and medical and logistic features of the victims may vary among different types of disasters (e.g., crushed patients in destructive disasters, gunshot or explosion-related injuries resulting from wars, drowned victims after flooding and tsunamis, widespread infection during pandemics).<sup>8</sup> Therefore preparations should be valid for all types of disasters (i.e., the “all-hazards approach,” which means managing different disasters according to a common plan)<sup>142-144</sup> but also must be complemented by a more focused disaster/disease-specific approach. This combination allows tailoring of the disaster response to the circumstances of each disaster and identifies the most pragmatic strategy for an efficient response.<sup>31,34,140,142-144</sup> Details regarding the “all-hazards approach” have been described elsewhere.<sup>142-144</sup> In this chapter, we focus on “renal disaster preparedness” strategies on a regional basis to minimize disaster-related risks for people with acute and chronic kidney diseases.

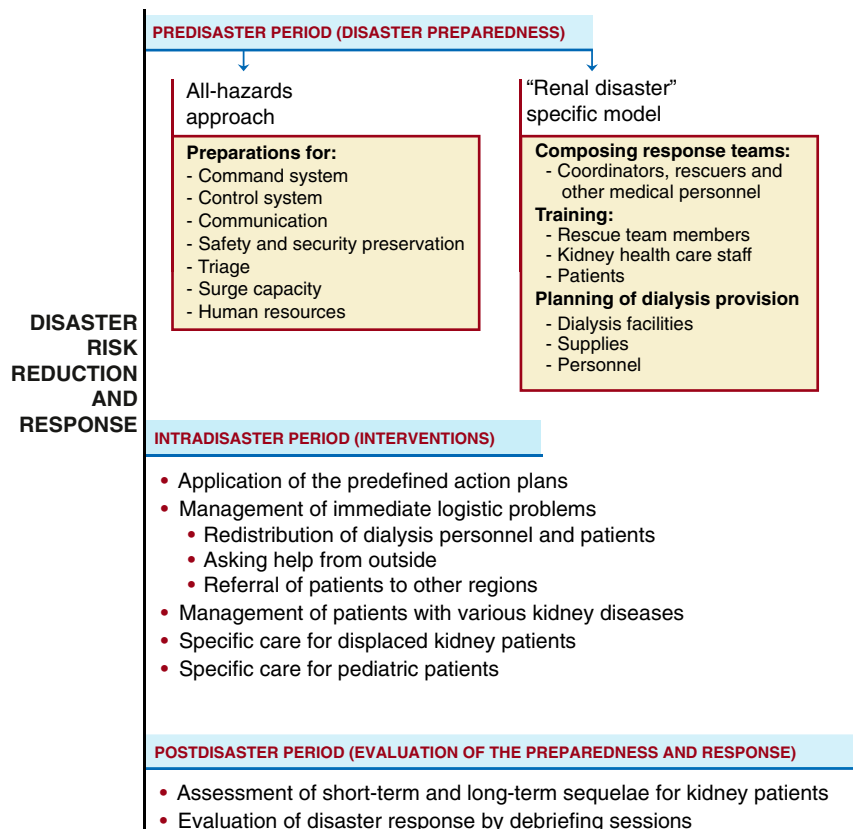
**RENAL DISASTER-FOCUSED PREPAREDNESS**

Preparing simple, clear, and pragmatic algorithms, which focus on particular disasters, is useful to avoid major medical and logistic errors due to panic, confusion, and inexperience during or in the early minutes after disasters (Figs. 82.5, 82.6, and 82.7). Overall, renal disaster preparedness encompasses several complex steps, summarized in Fig. 82.4.

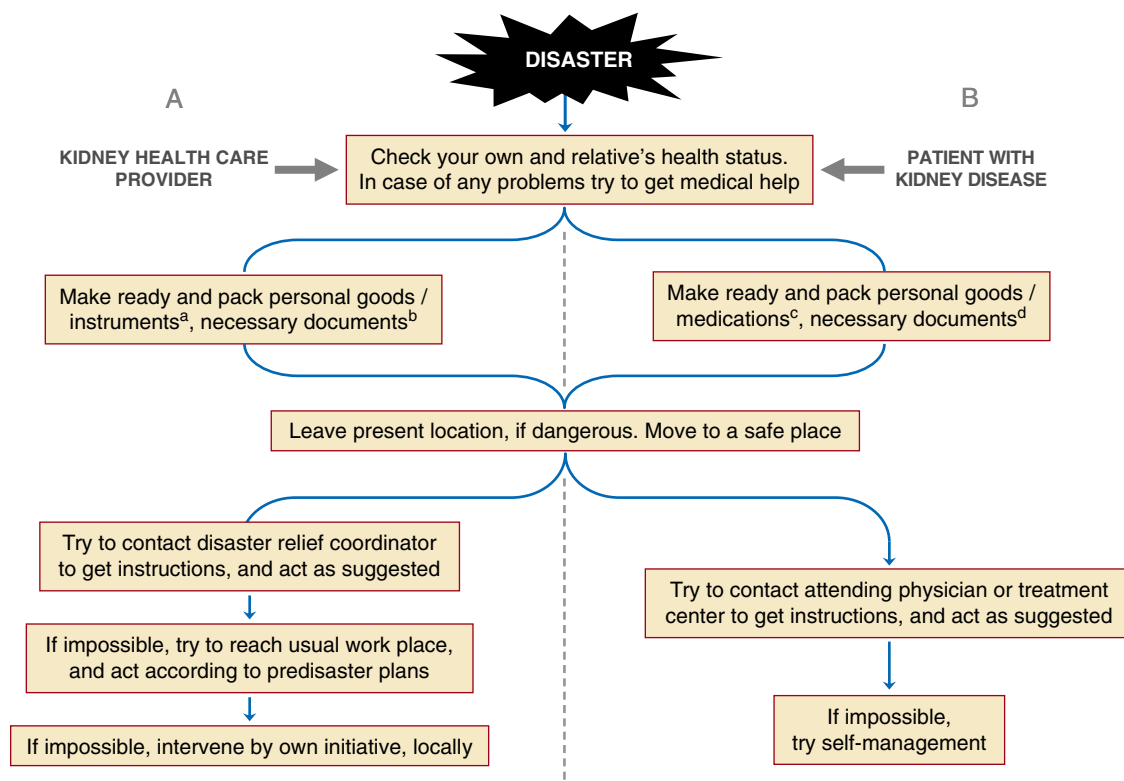
**COMPOSING DISASTER RESPONSE TEAMS**

Many stakeholders contribute to disaster response, and their commitment should already begin before disasters by generating intervention plans, continue during the acute phase of the disaster by implementing rescue and therapeutic interventions, and end after the disaster by rehabilitating the victims and evaluating the overall response.

Assigning trained and experienced disaster relief coordinators (DRCs) is crucial to achieve several tasks (Table 82.2).<sup>31,47,139,145</sup> A critical task of DRCs is preparing scenarios by dividing the country into sectors, each with an assigned local (sector) coordinator, to collaborate in conducting national/regional disaster response and support. Sector coordinators assess the working capacity of local personnel in their region, arrange personnel and material help for the affected region, and coordinate relocation of disaster



**Fig. 82.4 An overview of risk reduction, response, and retrospective evaluation of the response in various phases of disasters.** The all-hazards approach combined with the renal disaster-specific model in the predisaster period should be followed by application of predefined action plans and management of anticipated or instant/unexpected logistic problems. Management of patients with various problems related to kidney diseases, which are often even more severe in displaced kidney patients and in children, is a critical part of the intervention at this stage. In the postdisaster period, overall predisaster preparations and intradisaster response are evaluated, shortcomings in response are identified, and measures are defined to avoid repetition in future disasters.<sup>31,34,139-141</sup>



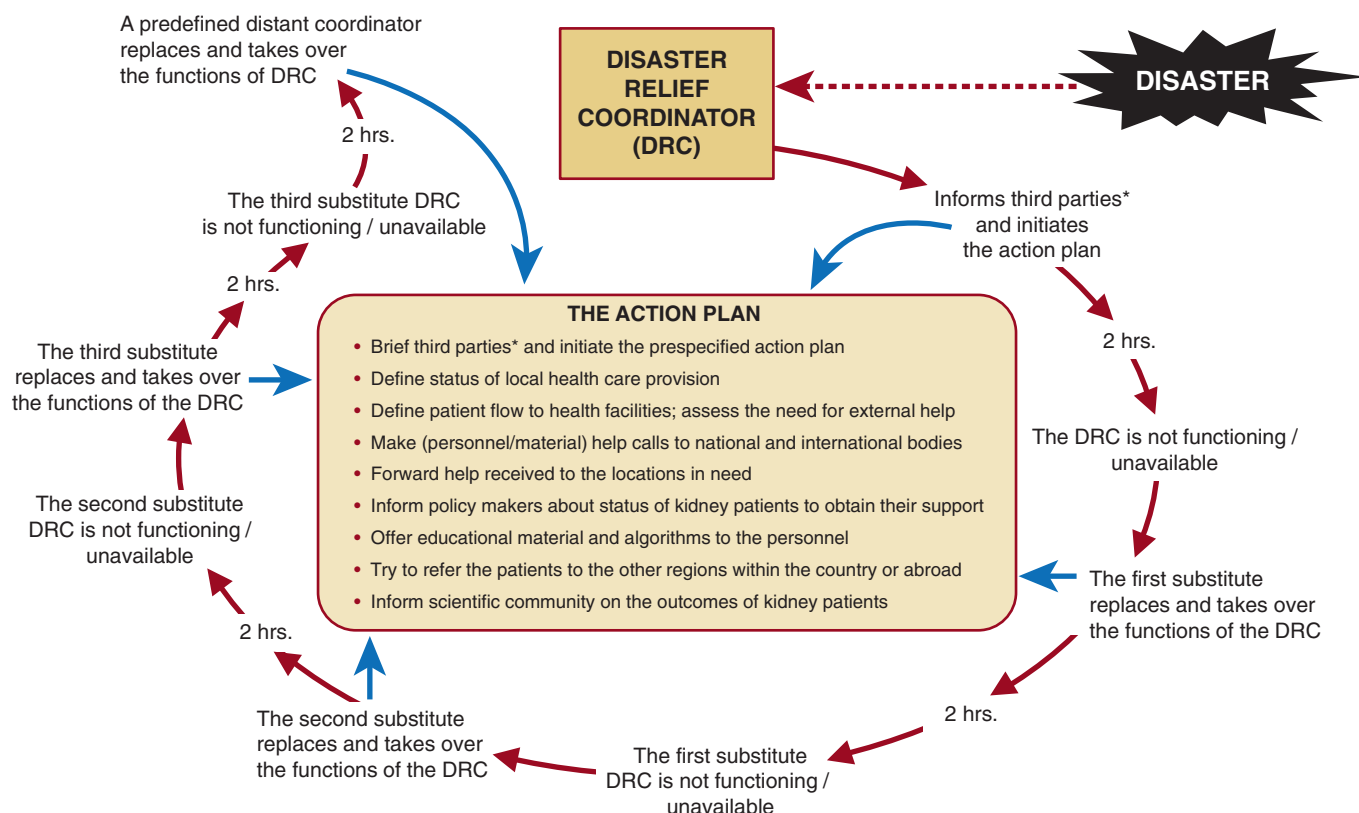
**Fig. 82.5 Example of a simple, clear, and informative algorithm to minimize chaos and panic immediately after experiencing a destructive disaster (e.g., earthquake, war, hurricane).** This algorithm describes early actions from the perspectives of kidney care providers (A) and patients with kidney diseases (B). Similar algorithms and/or infographics may (should) be created on various topics targeting the health care personnel or rescuers (e.g., interventions at the disaster field, issues to be considered for patient transport, indications for various medical interventions) or for the patients (e.g., how to secure themselves during the acute episode of a disaster, dietary suggestions, and simple instructions for self-treatment).

<sup>a</sup>Instruments: e.g., personal medical instruments kept at home.

<sup>b</sup>Necessary documents: e.g., action plans, treatment algorithms, and contact lists.

<sup>c</sup>Medications: e.g., immunosuppressants, antihypertensives, antidiabetics, and other critical medications.

<sup>d</sup>Necessary documents: e.g., medical reports, prescriptions, and treatment details.<sup>31,139</sup>



**Fig. 82.6 Renal disaster action plans for emergency response and organizing relief.** Within the first 2 hours of the disaster, the disaster relief coordinator (DRC) should brief third parties and initiate the action plan. If the DRC cannot function or is not reachable within the first 2 hours, the first local substitute will be automatically assigned. If the first local substitute does not respond or cannot be reached, the second local substitute will act as the DRC. This scenario is repeated unless even the third local substitute cannot be reached. In the latter case, a predefined distant coordinator (the fourth substitute) will take over the functions of the DRC. This pragmatic strategy has been effective in past disasters, particularly in the case of communication problems. Adaptations to the action plan considering the type and extent of the disaster, the status of the predisaster and postdisaster general and health care infrastructure of the affected region and country, and local circumstances may be necessary. (Adapted with permission from Sever MS, Lameire N, Vanholder R. Renal disaster relief: from theory to practice. *Nephrol Dial Transplant*. 2009;24(6):1730–1735.)

\*Third parties: authorities, substitute DRCs, the president of the local nephrology society and the nephrology community.

victims from the affected area for temporary rescue dialysis or maintenance treatment elsewhere in their region.<sup>131,139</sup>

Rescuers directly intervene in the disaster field, and kidney health care providers treat victims in health care facilities. Therefore selection, training, and assignment of these people according to personal skills and experience are essential to save as many lives as possible (Table 82.3).<sup>31,47,139,140,145</sup>

The most effective disaster response can be achieved by executing tasks that are specified by the DRCs; however, if these coordinators cannot be reached, rescue or medical personnel should take appropriate initiatives by themselves (see Fig. 82.5).<sup>31,139</sup>

## ORGANIZING TRAINING ACTIVITIES

Since all disasters are life-threatening, training of response team members to preserve their own security and that of their relatives should be the first step of preparation. Apart from that, training courses should target various tasks for different health care providers:

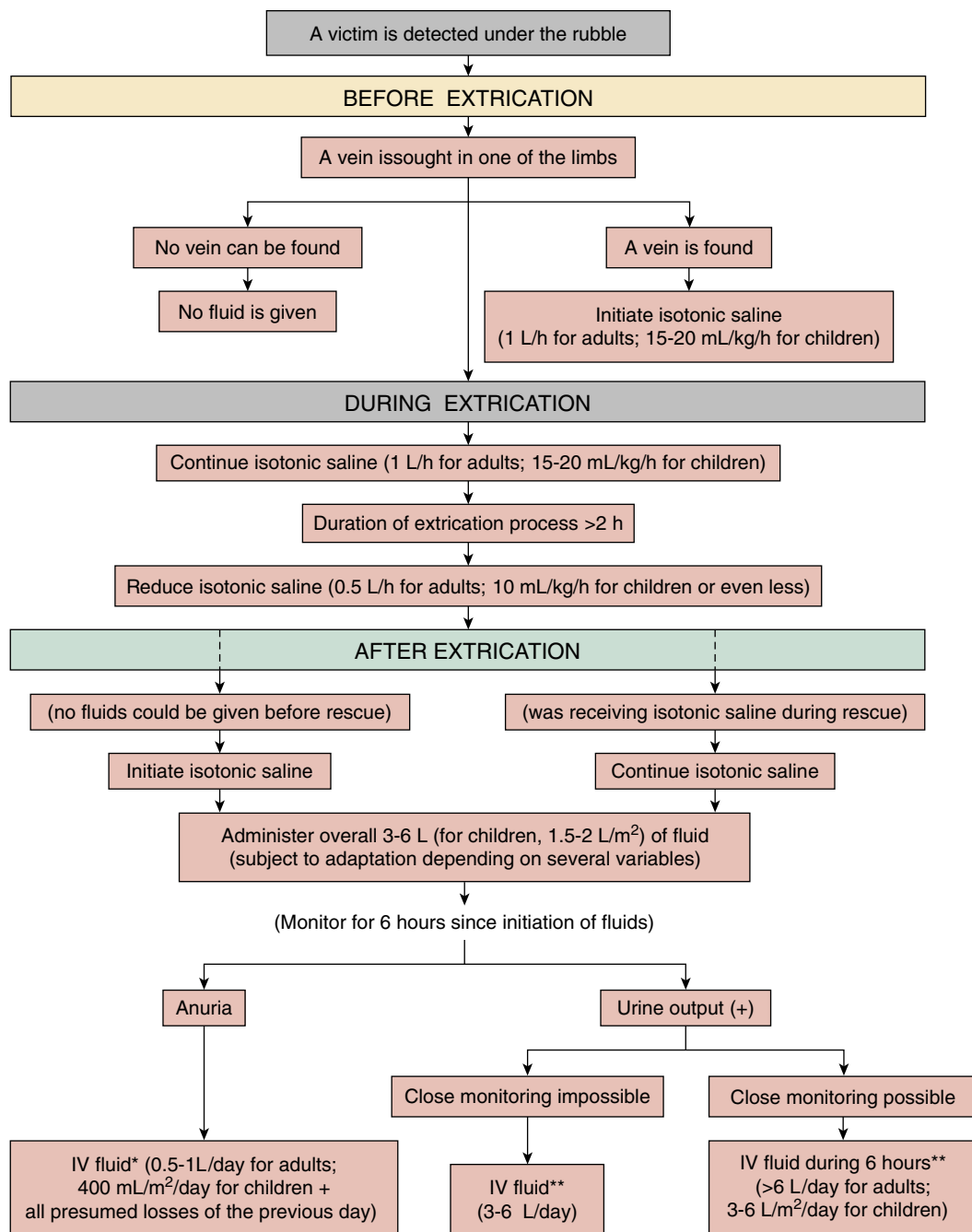
- *Rescue team members* should be trained on field interventions, triage practices, treatment of urgent problems on the spot and during transportation, and various other tasks (see Table 82.3). Overall, training courses should be adapted to anticipated disasters (i.e., first aid in gunshot injuries

in wars, extrication and preventive measures for crush syndrome in destructive disasters, and helping drowned victims during flooding).<sup>31</sup>

- Training courses for *kidney health care providers* should define early actions immediately after the disaster (see Fig. 82.5A), considering the disaster in focus. For example, dialysis nurses should be trained to not only dialyze injured victims but also disconnect patients from the machines and take other necessary actions (e.g., evacuation from the dialysis units, first aid for the victims, or triage practices) should a disaster occur during an ongoing dialysis session.<sup>31</sup>
- The first target in *patient education* should be to minimize panic reactions and describe the first response during the acute phase and early after destructive disasters (see Fig. 82.5B). Training sessions should also provide information on emergency patient preparedness plans, kidney-disease diet, and avoiding complications (e.g., how to avoid being injured during destructive disasters; contracting infections in pandemics and unhygienic disaster conditions), as well as self-management strategies if relevant medical personnel cannot be contacted (Table 82.4).<sup>31,47,139,145,146</sup>

Composing and updating a personal stock of medications and supplies (e.g., PD solutions and immunosuppressive





**Fig. 82.7 Fluid administration protocol to prevent crush-related AKI for entrapped victims of destructive disasters before, during, and after extrication.**<sup>155,156</sup> The intravenous route is preferred; however, if this cannot be achieved, intraosseous infusion may be considered as well, although this intervention may be impossible in chaotic disaster field conditions. If neither intravenous nor intraosseous accesses are possible, hypodermoclysis (subcutaneous infusion of isotonic fluids) at a rate of approximately 1 mL/min may be considered as a last resort. Up to 3 L/day of fluids can be given by this route; therefore it is not the ideal route for patients needing large volumes of fluids. Patients with skin or bleeding disorders or peripheral edema may not be appropriate candidates for hypodermoclysis.<sup>153</sup> Although oral rehydration is a possibility in alert patients with no risk of intraabdominal pathology and no need for imminent anesthesia,<sup>154</sup> exclusion of these probabilities may be problematic, especially in destructive disasters. (Modified with permission from Sever MS, Vanholder R. Management of crush victims in mass disasters: highlights from recently published recommendations. *Clin J Am Soc Nephrol*. 2013;8(2):328–335.)

\*Please see the text for advantages and disadvantages of various types of fluids.

\*\*Intensive fluid protocol is continued until disappearance of myoglobinuria (normalization of the urinary color), which usually occurs within 2 to 3 days following trauma; this fluid regimen is reduced or stopped in case of volume overload.<sup>34,157</sup>

medications) is crucial for patients. These supplies must be rotated regularly to prevent expiry and waste in the long term. Such simple measures have not yet been widely applied, even in well-developed, disaster-prone countries, such as Japan<sup>91</sup> or the United States.<sup>146,147</sup>

An acute disaster episode may occur when hemodialysis patients are connected to their dialysis machines. Due to panic and extreme workload, staff may not be able to disconnect patients immediately; therefore self-disconnection, either by “clamp and cut”<sup>148</sup> or “clamp and cap”<sup>149</sup> techniques, should be

**Table 82.2** Tasks and Responsibilities of Relief Coordinators to Reduce the Risks for Patients with Kidney Diseases During Various Phases of Disasters<sup>a,b; 31,47,139,145</sup>

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before | <ul style="list-style-type: none"><li>• Create disaster intervention scenarios and action plans, and divide country into several sectors</li><li>• Assign and train sector coordinators to intervene during disasters in their own sectors and to provide support to other sectors</li><li>• Create comprehensive algorithms to solve medical and logistic problems</li><li>• Organize and/or conduct translation of relevant guidelines</li><li>• Advocate to include disaster medicine courses in medical and paramedical curricula and congresses</li><li>• Contact international organizations for collaboration in case of future disasters</li></ul> |
| During | <ul style="list-style-type: none"><li>• Apply predefined measures to preserve own safety</li><li>• Initiate prespecified action plans (described in Fig. 82.6) immediately after the disaster</li><li>• Create solutions to solve instant logistic problems</li><li>• Coordinate management of patients with various kidney diseases</li><li>• Consider specific problems in children and displaced kidney patients</li></ul>                                                                                                                                                                                                                              |
| After  | <ul style="list-style-type: none"><li>• Follow ethical rules in all interventions</li><li>• Assess the efficiency of the disaster response and identify shortcomings for future improvements in future disasters</li><li>• Identify shortcomings in the treatment of patients with kidney diseases to avoid similar mistakes in the future</li><li>• Inform the scientific community on interventions and kidney patient outcomes</li></ul>                                                                                                                                                                                                                |

<sup>a</sup>Data adapted from REF.<sup>31</sup>

<sup>b</sup>This list is not exhaustive.

**Table 82.3** Tasks and Responsibilities of Rescuers and Kidney Health Care Providers to Reduce the Risks for Patients with Kidney Diseases Before, During, and After Disasters<sup>a,b; 31,47,139,140,145</sup>

|        |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before | <ul style="list-style-type: none"><li>• Acquire and provide training on disaster nephrology</li><li>• Disseminate relevant information to third parties</li><li>• Train staff and organize drills for management of disaster-related problems</li><li>• Train patients with kidney diseases on disaster preparedness and personal disaster response to minimize threats and drawbacks</li></ul>                       |
| During | <ul style="list-style-type: none"><li>• Apply predefined measures to preserve own safety</li><li>• Apply prespecified interventions, but make adaptations in case of problems or inefficiency</li><li>• Adapt treatment protocols for those injured or affected by the disaster in function of the logistic constraints</li><li>• Refer patients to higher-level hospitals if local treatment is impossible</li></ul> |
| After  | <ul style="list-style-type: none"><li>• Evaluate local disaster response, specify shortcomings and make adaptations for improvement</li><li>• Screen for- and treat long-term disaster-related comorbidities</li><li>• Inform the scientific community on interventions, and patient outcomes</li></ul>                                                                                                               |

<sup>a</sup>Data adapted from REF.<sup>31</sup>

<sup>b</sup>This list is not exhaustive.

**Table 82.4** Tasks and Responsibilities of Patients with Kidney Diseases to Reduce Risks Before, During, and After Disasters<sup>a,b; 31,47,139,145,146</sup>

|        |                                                                                                                                                                                                                                                                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before | <ul style="list-style-type: none"><li>• Acquire education and training on how to cope with disaster-related medical conditions</li><li>• Apply measures to decrease potential risks (e.g., maintaining a 2- week stock of medications and PD supplies)</li><li>• Participate in drills on management of disaster-related problems</li></ul> |
| During | <ul style="list-style-type: none"><li>• Apply predefined measures to preserve own safety</li><li>• Take predefined measures to minimize the health risks that are associated with disasters</li><li>• Apply self-management protocols if attending physicians or nurses cannot be contacted</li></ul>                                       |
| After  | <ul style="list-style-type: none"><li>• Be vigilant for any changes in health status</li><li>• Check status of own or alternative health care facilities for treatment possibilities</li></ul>                                                                                                                                              |

<sup>a</sup>Data adapted from REF.<sup>31</sup>

<sup>b</sup>This list is not exhaustive.

PD, Peritoneal dialysis.

taught to patients. A similar approach is valid for APD patients as well. Patients who have adequate knowledge and training must also be prepared to assist others.<sup>31</sup> Finally, the patients should be informed about alternative dialysis centers either inside or outside the affected region to relocate to when their usual center is no longer operational. Carrying an updated copy of their complete medical history, a list of medications, and treatment prescriptions is useful as electronic records from the original unit may not be readily available to the new center.

PLANNING OF DIALYSIS PROVISION

Planning of dialysis provision is extremely challenging because neither the number of patients who will need dialysis nor the status of dialysis facilities can be known in advance of the disaster; therefore all planning may be based on estimations. Such planning may be simpler for foreseeable disasters, such as hurricanes, flooding, and some wars<sup>31</sup> and “prophylactic dialysis” (i.e., extra dialysis sessions in advance of the disaster on top of the routine schedule for patients on maintenance hemodialysis may help decrease

**Table 82.5 Checklist for Information to Be Collected from Dialysis Units in the Context of Disaster Preparedness<sup>34</sup>**

- Name of the unit
- Private/public status
- Structural characteristics of the building
- Possibility for helicopter landing within a perimeter of 1 km
- Number and type of hemodialysis machines
- Normal staffing
- Number of nephrologists and other doctors and their contact information
- Availability of pediatric nephrologists
- Number of hemodialysis nurses and their contact information
- Number of hemodialysis technicians and their contact information
- Number of maintenance patients dialyzed in the unit; their name and contact information
- Maximum number of patients, who can be treated in that particular unit
- Availability of intensivists, intensive care positions, and possibilities to dialyze at the ICU

immediate postdisaster dialysis needs).<sup>96,150</sup> However, this strategy cannot be applied in unforeseeable disasters such as earthquakes, tsunamis, and terrorist attacks.

All nephrology and dialysis units in and around disaster-prone areas should develop their own detailed disaster preparedness plans to cope with sudden increases in dialysis needs. Specific information about these facilities should be registered in the regional/national preparedness plans (Table 82.5)<sup>34</sup> and databases that should be available to relevant stakeholders—coordinators, nephrologists, nurses, and patients.

Considering the possibility that many dialysis facilities will collapse, regional or national directories for nephrology and dialysis units should identify the capacities for material and/or personnel support, as well as accepting patients from nonfunctional units.<sup>34,107</sup> Likewise, considering the need for gigantic quantities of consumables for dialysis<sup>33</sup> and the potential for damage to warehouses that store dialysis supplies, international support will mostly be needed after massive disasters.<sup>40,140,141,151</sup> Therefore regional, national, and supranational preparedness should be considered in disaster-prone countries.<sup>17,34,140</sup>

Personnel planning is complex because of unpredictable shortages of staff members.<sup>8,139</sup> Many measures to cope with personnel shortage can be taken only after observing the extent of the disaster.

To summarize, preparedness plans for the interventions during the disaster should be detailed and solve the questions “who,” “what,” “when,” and “how” clearly.<sup>139</sup> Regular drills are needed to compensate for potential inexperience of the responders.

### INTRADISASTER PERIOD (DISASTER RESPONSE)

Overall principles of disaster response are similar for all stakeholders (i.e., DRCs, health care providers, and

patients), which entails initially checking one’s own and relatives’ health status, leaving the current location to move to a safer place, and applying previously defined action plans (see Fig. 82.5).

### IMPLEMENTATION OF THE PREDEFINED ACTION PLANS

Renal disaster response is initiated by the DRC, who applies predefined action plans. If during destructive disasters the DRC becomes disabled for any reason, he or she is replaced by a prespecified substitute to take over his or her position for applying the action plans (see Table 82.2; Fig. 82.6). Even if overall principles apply to any disaster, there may be a need for adaptations according to the type and dimensions of the disaster.

### MANAGEMENT OF IMMEDIATE LOGISTIC PROBLEMS

These interventions are mostly focused on dialysis provision and may include but are not limited to. 1. Reassigning personnel from nonfunctioning to functioning units, which may reduce personnel shortages.<sup>34</sup> Alternatively, working hours of personnel may need to be increased, however, with an increased risk of burnout and malpractice.<sup>48</sup> 2. In case of a nonfunctioning dialysis unit, support from other units should be planned (e.g., by relocating the patients to surrounding functional units).<sup>34</sup> 3. Referral of the maintenance in-hospital hemodialysis patients to nearby satellite units is useful to make space available in the hospital dialysis units for complex and acute patients, who may require tertiary hospitals and/or ICU admission.<sup>17</sup> 4. Requests for material and/or dialysis personnel support from outside the disaster area or country have been useful.<sup>140,145,152</sup> However, these campaigns must be organized by coordinators with logistics experience and based on correct information from the affected area; otherwise, external help may be harmful.<sup>140,152</sup> Also, cost, politics, and logistics may be important barriers.

Evacuation of patients to other regions or even abroad may be considered if extensive damage makes local dialysis impossible.<sup>31,35,41-43,107</sup> However, this option is complicated by the challenges of long-distance patient relocation.

### MANAGEMENT OF PATIENTS WITH VARIOUS KIDNEY DISEASES

Management strategies may differ according to the type and severity of the kidney disease, as well as the nature of the disaster. Therefore “to do list” priorities for the physicians (ideally nephrologists) may change (Table 82.6).<sup>34</sup>

#### PATIENTS WITH ACUTE KIDNEY INJURY

Specific management of patients with AKI depends on the underlying etiology. Considering its high incidence, management of crush injury-related AKI is discussed in this chapter; whereas treatment of other etiologies-related AKI is described in Chapters 28 and 64.

Hypovolemia, the main pathogenetic mechanism in crush-related AKI, can be prevented by timely administration of appropriate fluids (Fig. 82.7).<sup>34,59,153-157</sup> Fluid volumes should be individualized, considering the time spent under

**Table 82.6 “To-Do List” for Nephrologists/Physicians for Treatment of Patients With Acute Kidney Injury During Various Phases of Destructive Disasters<sup>a</sup>**

|                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial phase<br>(interventions on<br>the spot)                                                         | <ul style="list-style-type: none"><li>• Detect, rescue, and triage the victims, and perform a primary survey</li><li>• Stop bleeding (if any), maintain airway patency, and apply spine immobilization (if needed)</li><li>• Treat any other life-threatening medical complications<sup>b</sup></li><li>• Start fluid administration with consideration of the specific conditions of the patient and disaster (Fig. 82.7)</li><li>• Organize urgent transport to the nearest and most appropriate hospital, with appropriate measures when needed<sup>c</sup></li></ul> |
| At admission to<br>hospitals<br>(interventions<br>at emergency<br>units of<br>reference<br>hospitals)   | <ul style="list-style-type: none"><li>• Register (if possible with the ID), check vital signs, and reperform triage</li><li>• Check type of fluids, and determine urgent electrolyte levels; if present, immediately treat hyperkalemia</li><li>• Evaluate indications for referral to the ICU</li><li>• After excluding urethral bleeding, insert a bladder catheter</li><li>• Consult with relevant specialties</li></ul>                                                                                                                                              |
| Clinical course<br>(interventions<br>during the<br>clinical<br>course at the<br>reference<br>hospitals) | <ul style="list-style-type: none"><li>• Make frequent visits</li><li>• Avoid nephrotoxic medications</li><li>• Evaluate evolution, and treat complications accordingly</li><li>• Apply the most appropriate dialysis modality considering medical and logistic features (Table 82.7)</li><li>• Identify the reason(s) of unexpected findings or unexpected alterations in the clinical course</li></ul>                                                                                                                                                                  |

<sup>a</sup>Although slight modifications may be required, overall, these principles are valid for the entire spectrum of patients with kidney diseases and in nondestructive disasters (e.g., pandemics as well). Actions needed at an earlier stage (i.e., before medical intervention) are visualized in Fig. 82.5A.

<sup>b</sup>May include but not limited to respiratory failure, hypotension/shock, hypertension, myocardial ischemia, and infarction, left ventricular failure.

<sup>c</sup>“Appropriate measures” refers to initiating or continuing full spinal immobilization for patients with spinal trauma, initiating or continuing life-saving measures for seriously injured victims, and organizing accompaniment by medical personnel, if needed. ICU, Intensive care unit; ID, “identity” or “identification.”

the rubble; the length of the extrication procedure; age, volume status, urine flow, and trauma pattern of the victim; environmental conditions, essentially ambient temperatures; and the ability or not to closely monitor the patient.<sup>34,57,155,158</sup>

Regarding the type of fluid, isotonic saline is preferred because of its effectiveness for volume replacement, ready availability, and low risk of side effects. Bicarbonate

solutions (prepared by adding 50 mEq sodium bicarbonate to 1 L half-isotonic saline) may correct metabolic acidosis, decrease the risk of hyperkalemia, and prevent intratubular myoglobin and uric acid cast formation, but they are usually not available in chaotic disaster conditions.<sup>34</sup> The efficacy of mannitol is controversial,<sup>159,160</sup> and blind or unjustified usage of potassium-containing solutions (e.g., Ringer lactate) is contraindicated in patients with crush injuries<sup>161,162</sup> because of the high risk of hyperkalemia in crush syndrome.

Patients should be dialyzed by standard criteria, although more liberal indications (even prophylactic dialysis) and more frequent sessions should be considered in crush-related AKI with the anticipation of potential complications, especially fatal hyperkalemia.<sup>33,34,163,164</sup> Intermittent hemodialysis (IHD) offers medical and logistic advantages over continuous renal replacement therapy (CRRT) or PD (Table 82.7).<sup>33,34,54</sup>

In addition to dialysis, many other therapeutic interventions may be indicated (e.g., fasciotomies,<sup>165,166</sup> amputations,<sup>167,168</sup> hyperbaric oxygen administration,<sup>169,170</sup> and blood transfusions).<sup>80,171</sup> These interventions are highly controversial, however, as they may be associated with significant medical and logistic benefits<sup>34,56,80,172,173</sup> and harms.<sup>34,55,158,165,174-176</sup> Discussing technical details of these interventions is outside the scope of this chapter.

#### PATIENTS WITH CHRONIC KIDNEY DISEASE NOT ON KIDNEY REPLACEMENT THERAPY

Optimal management of underlying diseases (e.g., diabetes, hypertension, or glomerulonephritis) is mandatory to reduce progression of CKD, although this is hard to achieve especially in protracted disasters.<sup>31,177</sup> Replacement of in-person visits by telemedicine technology may minimize the need for transportation and other risks of clinic attendance.<sup>31,66</sup> Problems in obtaining medications and appropriate diet may be mitigated by creating a personal stock of these items in advance.<sup>31</sup> Decision to modify immunosuppressive treatment for primary disease management should be individualized considering local circumstances and patient factors.<sup>66</sup>

#### PATIENTS ON HEMODIALYSIS

The disparity between health care demand and supply of hemodialysis during disasters<sup>35,178</sup> can be mitigated either by decreasing the demand or increasing the operational capacity.

Decreasing demand can be achieved by less frequent dialysis (e.g., twice- or even once-weekly) dialysis.<sup>35</sup> This strategy may also be useful to decrease the risk of contracting the disease during pandemics;<sup>179,180</sup> however, such strategy may be associated with hypertensive crises, volume overload, and hyperkalemia.<sup>179</sup> Risks are lower in older patients, in those with lower ultrafiltration requirements and/or with normal predialysis potassium and phosphate levels.<sup>179</sup> Less frequent dialysis may be inappropriate for patients with high risk of hyperkalemia, such as those with crush injuries or requiring renin-angiotensin-aldosterone system inhibitors.

Increasing dialysis capacity may be possible by shortening sessions in stable patients to 3 hours, allowing more dialysis sessions per machine, per day.<sup>181</sup> However, this may result in



**Table 82.7 Medical and Logistic Advantages and Drawbacks of Various Dialysis Modalities in Disaster Victims With Crush-Related Acute Kidney Injury<sup>33,34,54</sup>**

|      | Medical Advantages                                                                                                                                                                            | Medical Drawbacks                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IHD  | <ul style="list-style-type: none"> <li>• High clearance rate of low-molecular-weight solutes, like urea and potassium</li> <li>• Possibility for dialyzing without anticoagulation</li> </ul> | <ul style="list-style-type: none"> <li>• Priming volume may induce hypotension</li> <li>• High risk of disequilibrium syndrome</li> </ul>                                                                                                 |
| CRRT | <ul style="list-style-type: none"> <li>• Better volume control</li> <li>• Low risk of disequilibrium syndrome</li> <li>• Possibility for higher calorie administration</li> </ul>             | <ul style="list-style-type: none"> <li>• Need for continuous heparinization</li> <li>• Low removal capacity for small solutes, like potassium</li> </ul>                                                                                  |
| PD   | <ul style="list-style-type: none"> <li>• No need for vascular access</li> <li>• Less hemodynamic instability</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>• Low clearance rate</li> <li>• Difficulty to maintain sterile conditions</li> <li>• Difficult application in some patients, especially those with severe thoracic and abdominal trauma</li> </ul> |
|      | Logistic Advantages                                                                                                                                                                           | Logistic Drawbacks                                                                                                                                                                                                                        |
| IHD  | <ul style="list-style-type: none"> <li>• Possibility to treat several patients per day in the same position</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>• Need for experienced personnel and technical assistance</li> </ul>                                                                                                                               |
| CRRT | <ul style="list-style-type: none"> <li>• For CAVH application, no need for electricity and only minimal equipment is needed</li> </ul>                                                        | <ul style="list-style-type: none"> <li>• Treatment possibility for only one patient, per machine per day</li> <li>• Problems with transportation of fluid bags</li> </ul>                                                                 |
| PD   | <ul style="list-style-type: none"> <li>• For CAPD, no need for water and electricity</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>• Problems with transportation of fluid bags</li> </ul>                                                                                                                                            |

*CAPD*, Continuous ambulatory peritoneal dialysis; *CAVH*, continuous arteriovenous hemodiafiltration; *CRRT*, continuous renal replacement therapy; *IHD*, intermittent hemodialysis; *PD*, peritoneal dialysis.

underdialysis, especially if vascular access is suboptimal, with consequent low blood flow rates or recirculation. Material supplies must be sufficient to enable the extra dialyses.

During disasters, patients on home hemodialysis have a lower risk for disaster-related problems due to a less need for transport and involvement of expert personnel, and also less risk of disease transmission during pandemics.<sup>182</sup>

The newer compact dialysis systems consume less dialysis water but remain dependent on availability of either tap water or sterile fluid in bags, which may be problematic during disasters. Dialysate regeneration systems are under development or in the test phase<sup>183,184</sup>; and the availability of these should also drastically reduce water needs, perfecting the old concept of the Redy system.<sup>55,185</sup>

If local dialysis treatment is not possible, and patients cannot be transferred to other regions or countries, switching HD patients to acute PD is an option to avoid emergent complications.<sup>101,186</sup>

For each scenario, optimizing medical management to avoid complications through use of diuretics, potassium-lowering agents, cathartics, strict dietary measures, and fluid restriction is mandatory.<sup>16,66</sup>

## PATIENTS ON PERITONEAL DIALYSIS

During disasters, continuous ambulatory peritoneal dialysis (CAPD) may be more practical than hemodialysis, because it obviates the requirement for transportation to dialysis units and does not depend on tap water, electricity, or dedicated personnel.<sup>107</sup> APD is even more appropriate than CAPD as it lowers the risk of peritonitis and increases efficiency; however, by its dependency on electricity, power outages may create problems, unless batteries/generators are available. When appropriate catheters and/or fluids cannot be found, acute catheters improvised from sterile medical

tubing (e.g., nasogastric tubes or central vein catheters) and dialysate generated from 0.45% saline or Ringer lactate may be used temporarily.<sup>101,187,188</sup> In case of a shortage of PD solutions, reducing the number of exchanges should be considered as well.<sup>31,66</sup>

Intensive training of the patients and families is important to optimize PD practice in unhygienic conditions. If the environment is not suitable, disconnections may be postponed until circumstances improve.<sup>189</sup> Training courses should target prevention of peritonitis, exit-site and catheter infections, and self-treatment of complications.<sup>101,129</sup> Assembling a disaster kit and having the contact information of dialysis suppliers may be useful in case of delivery problems.<sup>31,190</sup> If continuation of PD is not possible, applying strict diet and medical management of complications are mandatory, as well as switching patients to HD or hybrid treatments, if possible. When all measures fail, the patients must be evacuated outside the disaster area.

## KIDNEY TRANSPLANT RECIPIENTS

After massive disasters, postponement of new transplantation activity is sometimes considered. Nevertheless, such a life-saving therapy should not be slowed down if local operational resources permit continuing.<sup>108</sup>

Lack of immunosuppressants can be solved by adapting therapies (e.g., switching from triple to double immunosuppressive drug regimens and targeting increased dosages with higher blood levels).<sup>191,192</sup> However, this strategy may elicit rejections and/or drug toxicity.<sup>31</sup>

Training of the transplant recipients on diet, self-treatment of mild complications, and infection prophylaxis are vital, especially if people living with a functioning transplant are forced after destructive disasters to live in crowded shelters. Education on empirical usage of antibiotics (when available)

is a possible option, if dedicated experts are unavailable. If logistic problems or complications (e.g., transplant rejection) cannot be solved locally, the patients should urgently be evacuated to other regions or countries.<sup>109</sup>

The frequency of in-person hospital visits may be decreased during the first 3 months post-transplantation. If, after the third month, blood testing can be performed conveniently at a different location than at the transplantation center, follow-up by means of telemedicine technology may obviate the need for in-person hospital visits.<sup>131,193,194</sup> However, this policy may increase the risk of overlooking complications by lack of physical examination and direct contact with a physician.

### SPECIFIC CARE FOR DISPLACED PATIENTS WITH KIDNEY DISEASES

Displacement may be associated with interruptions in regular diet and medical treatment, missed dialysis sessions, and frequent hospitalizations; therefore preparative measures are mandatory.<sup>95,122,195</sup> In addition to the general measures described earlier, some extra considerations are needed for displaced patients with kidney diseases<sup>66,101,109,119,122,195,196</sup>:

1. Most importantly, patient education is required to avoid panicky displacement, which is much more risky than organized evacuation.
2. Adherence to predefined dietary measures and bringing along a stock of critical medications when traveling may avoid problems.
3. Considering their high vulnerability, evacuation and relocation of dialysis and transplanted patients should be prioritized during massive displacement.
4. Evacuation should be planned in stages, with the possibility to dialyze at intermediate locations, especially in case of relocation over long distances.
5. Nutrition, health, and hygiene during evacuation should be ensured.
6. The patients with kidney diseases should be met at the border of the country of destination, and immediate triage should be performed to identify those in need of emergency intervention.

### SPECIFIC CARE FOR PEDIATRIC PATIENTS

When preparing disaster plans, features of pediatric nephrology units should be included among national directories for nephrology and dialysis facilities. Planning is vital for children, especially for patients on kidney replacement therapy. Training and preparations should target not only pediatric nephrology centers but also patients and their families or caregivers.<sup>131</sup>

Overall, principles for prevention and treatment of AKI in children during disasters are similar to those for adult patients. A fluid resuscitation protocol for prevention of crush-related AKI in children is included in Fig. 82.7.<sup>34,59,153-157</sup>

Shortage of pediatric dialysis services is frequent after massive disasters<sup>43</sup> and can be improved by increasing dialysis efficacy; that is, by using the upper limits of dialyzer surface area (1–1.2 times of body surface area) and dialyzer blood flow rate (5–8 mL/min/kg).<sup>197</sup> In hemodialysis units, if there is a concern about water purity, if possible, low-flux dialyzers should be preferred over high-flux filters.<sup>131</sup>

Health care staff with experience in pediatric dialysis and usage of pediatric-sized dialysis equipment are scarce even under normal conditions<sup>198</sup>; this is exacerbated in disaster circumstances.<sup>19</sup> Therefore personnel with previous pediatric patient care experience, even if retired, may be

helpful. Considering the possible unavailability of primary caregivers during disasters, as many family members as possible should be educated on taking responsibility, if needed.

Special attention should be paid to the storage of different-sized catheters, dialyzers, and PD fluid bags for children, which are different to those used in adults.<sup>131</sup> In case of shortage, use of PD fluid bags should be maximized. As 2 to 3 L bags available for adults are also used in children, who need lower fill volumes, the same bag may be used for several consecutive exchanges. CAPD can be continued without disconnecting the bag as long as there is enough fresh dialysate volume available to adequately refill the peritoneal cavity. This strategy may also reduce the risk of infection by reducing the number of connections.<sup>189</sup> In patients on APD, and when fluids are scarce, the dialysis time may be prolonged to take optimal profit from the solutions. If problems persist, other treatment centers should be contacted to obtain missing solutions from them. Since usage of APD may be impossible due to power outages, all potential caregivers of pediatric PD patients should be trained in CAPD.<sup>103</sup> Support for material and personnel help from nongovernmental organizations and patient groups may be helpful to safeguard the continuity of PD supplies during disasters.

Due to the different physiognomy and needs of each individual child, children on HD or PD require highly individualized prescriptions. It is imperative that medical reports with information on current treatment protocols and medications are available in case of evacuation to a new health care center.

## POSTDISASTER PERIOD (EVALUATION OF THE RESPONSE)

During disasters, limited treatment possibilities enhance the risk of progression and complications of CKD<sup>86,199</sup> and contribute to the progression of AKI to CKD.<sup>84,85</sup> Also, nonurgent or silent medical problems might have been postponed, overlooked, disregarded or neglected for security, logistic or economic reasons.<sup>66</sup> Therefore patients with acute or chronic kidney diseases who have experienced a disaster should be screened and evaluated for long-term sequelae in the postdisaster phase.<sup>66</sup> Since all risks are more pronounced after long-lasting disasters, screening should, by preference, start as soon as possible.<sup>31</sup>

Unfortunately, natural and man-made disasters frequently recur; therefore once the situation is under control, an in-depth analysis of the disaster response by all stakeholders should take place in debriefing meetings, and deficiencies in the response should be identified. This strategy may provide the basis for preparing for future disasters because learning from experience is indispensable to avoid repeated mistakes.

## ETHICAL ISSUES IN PROVIDING KIDNEY HEALTH CARE DURING DISASTERS

The classical ethical principles of nonmaleficence (avoiding harm), beneficence (doing good), distributive justice, and respect for autonomy uniformly apply in disaster

conditions as well. However, some principles of medical ethics may necessitate different approaches depending on the severity of the event, collective needs, degree of preparedness, availability of resources, and cultural and social factors.<sup>31,200,201</sup>

In view of the disparity between health care demand and supply, the main ethical concern is fair distribution of limited medical supplies and manpower. A utilitarian approach, which often necessitates triage, has traditionally been used to maximizing benefit and saving as many lives as possible.<sup>200,202,203</sup> Considering that only healthy and functioning staff can help the victims, priority of care should be given to health care personnel.<sup>108,202,203</sup> Next, treatment of the sickest but salvageable patients should be prioritized. When patients in need of the same treatment have comparable medical and logistic risks and prognoses, younger age may be prioritized for life-saving therapies including dialysis,<sup>203-205</sup> although clinical frailty, survival probability, or life expectancy do not necessarily correlate with age.<sup>204</sup> Other triage criteria may be the “first-come, first-served” principle, random selection, or even a lottery.<sup>31,66,203,206,207</sup> Transparency in communication with the public about triage strategies may reduce conflicts at the bedside.<sup>208</sup> If basic principles and values are conflicting with disaster circumstances, physicians may be compelled to apply their practical judgment to decide how to act.<sup>201</sup>

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